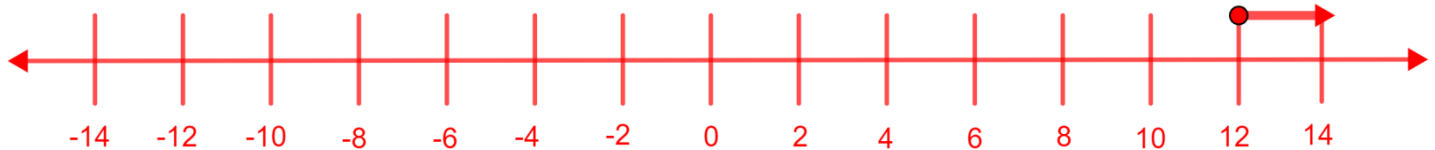


According to Florida Building Code, a parking space must be no less than 12 feet wide.

1. Write an inequality to represent the width, w , that parking spaces can be.

$$w \geq 12$$

2. Represent the inequality on a number line.



3. List two examples of possible widths, in feet, of the parking spaces.

Answers can vary. 12 ft, 14 ft.

4. Give one example of a width that is true for the inequality but does not make sense in the real world.

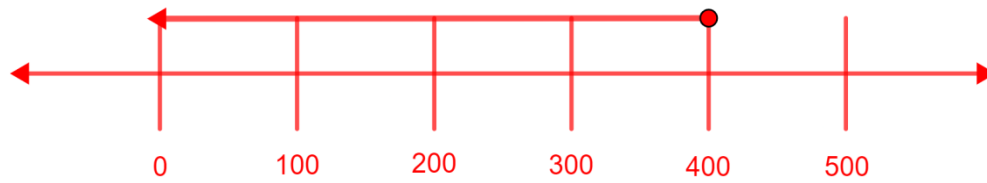
Answers can vary. 1,000 ft.

The maximum weight for a rider on a collapsible-floor water slide is 400 pounds.

5. If w represents the weight of the rider, write an inequality to describe all possible values of w .

$$w \leq 400$$

6. Represent the inequality on a number line.



7. Does the inequality that represents the situation include 0 as a possible weight? Why or why not?

Yes, any weight less than or equal to 400 pounds is a possible solution including zero in terms of the inequality.

8. Explain why a negative number does not make sense in this context.

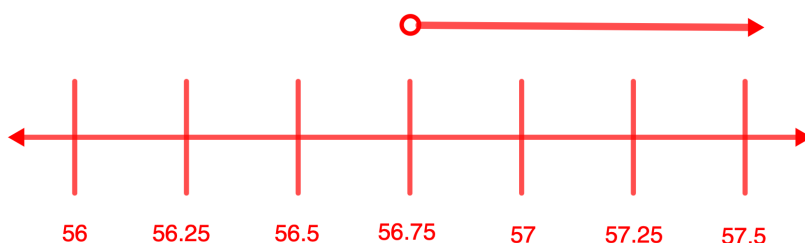
A negative number does not make sense in this context because riders will not have a negative weight.

A farmer plans to plant more than 56.75 acres of lemons.

9. If L represents the number of acres of lemons the farmer plants, write an inequality that represents the number of acres the farmer plants.

$$L > 56.75$$

10. Represent the inequality on a number line.



Grade 6 Accelerated
Unit 14
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

Solve.

$$\begin{aligned} 1. \quad x - 14.5 &= 41.7 \\ &\quad +14.5 \quad +14.5 \\ x &= 56.2 \end{aligned}$$

$$\begin{aligned} 2. \quad 6 \cdot \frac{k}{6} &= 12.25 \quad \cdot 6 \\ k &= 73.5 \end{aligned}$$

$$\begin{aligned} 3. \quad 0.75 - p &= 0.19 \\ -1p &= 0.75 - 0.19 \\ -1p &= 0.56 \\ p &= 0.56 \end{aligned}$$

$$\begin{aligned} 4. \quad \underline{18} &= \underline{0.5t} \\ 0.5 \quad 0.5 \\ t &= 36 \end{aligned}$$

$$\begin{aligned} 5. \quad 6.5 + f &= 17.11 \\ -6.5 \quad -6.5 \\ f &= 10.61 \end{aligned}$$

$$\begin{aligned} 6. \quad \frac{3}{2} \cdot \frac{2}{3}r &= 22.8 \quad \cdot \frac{3}{2} \\ r &= 34.2 \end{aligned}$$

7. Mike made jelly and filled three equal sized jars with $3\frac{2}{3}$ cups of jelly. How much jelly did Mike make?

$$x \div 3\frac{2}{3} = 3$$

Mike made 11 cups of jelly.

$$x = 3 \cdot 3\frac{2}{3}$$

$$x = 3 \cdot \frac{11}{3} = 11$$

8. Davis cut $\frac{1}{6}$ feet off of a piece of fabric, leaving $4\frac{1}{2}$ feet of fabric left. How long was the fabric before Davis cut it?

$$x - \frac{1}{6} = 4\frac{1}{2}$$

The fabric was originally $4\frac{2}{3}$ ft. long.

$$x = 4\frac{1}{2} + \frac{1}{6} = 4\frac{2}{3}$$

9. An environmental science teacher has a 10-gallon fish tank. Currently, the tank has $4\frac{9}{11}$ gallons of water inside. How much more water is needed to fill the tank?

$$10 - x = 4\frac{9}{11}$$

$5\frac{2}{11}$ gallons are needed to fill the tank.

$$x = 9\frac{11}{11} - 4\frac{9}{11}$$

$$x = 5\frac{2}{11}$$

10. The 6th grade class at a school is volunteering to decorate a hallway for homecoming. The students have decorated $\frac{5}{16}$ of a mile in decorations which is $\frac{3}{4}$ of the hallway. What is the length of the hallway being decorated?

$$\frac{4}{3} \cdot \frac{3}{4}x = \frac{5}{16} \cdot \frac{4}{3}$$

The hallway is $\frac{5}{12}$ of a mile long.

$$x = \frac{5}{12}$$

Grade 6 Accelerated
Unit 14
Lesson 3 Homework

Name Answer Key
Date _____
Period _____

1. What is the inverse of addition?

subtraction

2. What is the inverse of subtraction?

addition

For questions 3-7, use the algebra tiles shown.



3. What expression represents the left side of the equation?

-9

4. What expression represents the right side of the equation?

$-7 + x$

5. Write the algebraic representation of the equation.

$-9 = -7 + x$

6. What operation would be used to solve the equation?

Subtracting negative 7 or adding 7

$$7. \quad y + 5 = 17$$

$$\quad \quad -5 \quad -5$$

$$y = 12$$

$$8. \quad -8 + x = 66$$

$$\quad \quad +8 \quad +8$$

$$x = 74$$

$$9. \quad -6 + x = -10$$

$$\quad \quad +6 \quad +6$$

$$x = -4$$

$$10. \quad 14 + x = 18$$

$$\quad \quad -14 \quad -14$$

$$x = 4$$

Grade 6 Accelerated
Unit 14
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

Solve.

$$\begin{array}{rcl} 1. & 17 = 10 + x & \\ & -10 & -10 \\ & \hline & x = 7 & \end{array}$$

$$\begin{array}{rcl} 2. & x - 17 = 231 & \\ & +17 & +17 \\ & \hline & x = 248 & \end{array}$$

$$\begin{array}{rcl} 3. & -229 = -20 + h & \\ & +20 & +20 \\ & \hline & h = -209 & \end{array}$$

$$\begin{array}{rcl} 4. & 313 = g - 91 & \\ & +91 & +91 \\ & \hline & g = 404 & \end{array}$$

$$\begin{array}{rcl} 5. & y - 6 = -19 & \\ & +6 & +6 \\ & \hline & y = -13 & \end{array}$$

$$6. \quad \begin{array}{rcl} 18 + u & = & -3 \\ -18 & & -18 \end{array}$$

$$u = -21$$

$$7. \quad \begin{array}{rcl} y + 9 & = & 31 \\ -9 & & -9 \end{array}$$

$$y = 22$$

$$8. \quad \begin{array}{rcl} 8 + x & = & -77 \\ -8 & & -8 \end{array}$$

$$x = -85$$

$$9. \quad \begin{array}{rcl} -13 + x & = & 117 \\ +13 & & +13 \end{array}$$

$$x = 130$$

$$10. \quad \begin{array}{rcl} 41 + x & = & 809 \\ -41 & & -41 \end{array}$$

$$x = 768$$

Grade 6 Accelerated
Unit 14
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

Write and solve an algebraic equation for each of the following.

1. Hiro has 18 highlighters, which is 4 fewer than Martin has. How many markers does Martin have?

$$\begin{array}{rcl} m - 4 & = & 18 \\ +4 & +4 & \\ \hline m & = & 22 \end{array}$$

Martin has 22 highlighters.

2. A college engineering program receives 1,245 applications this year, of which only 200 can be accepted. How many students are not accepted?

$$\begin{array}{rcl} a + 200 & = & 1245 \\ -200 & -200 & \\ \hline a & = & 1045 \end{array}$$

1,045 are not accepted.

3. Pepper has an 8 pack of markers. She let Shawn borrow some of her markers from the box. When she counted the markers in the box again, she had 5 remaining. How many markers did Shawn borrow?

$$\begin{array}{rcl} 8 - m & = & 5 \\ \hline m & = & 3 \end{array}$$

Shawn borrowed 3 markers.

4. Mrs. Tuttle took her class to the media center for their class photo. Of her 18-student class, only 13 were present for picture day. How many students were absent?

$$\begin{array}{rcl} 18 - a & = & 13 \\ \hline a & = & 5 \end{array}$$

5 students were absent.

5. Jasmine consumed a total of 1,450 calories of her 2,000 daily calorie limit. How many more calories can Jasmine consume?

$$\begin{array}{rcl} 1450 + c & = & 2000 \\ -1450 & -1450 & \\ \hline c & = & 550 \end{array}$$

Jasmine can consume 550 more calories.

6. A grocery store set out 78 apples on a display stand. At the end of the day, there were 31 apples remaining. How many apples were sold?

$$78 - a = 31 \quad 47 \text{ apples were sold.}$$

$$a = 47$$

7. A soccer team is doing a car wash to raise money. The team earned 395.45 between cars and donations. If \$176.40 was raised from washing cars, how much was made from donations?

$$176.40 + d = 395.45 \quad \$219.05 \text{ was made from donations.}$$

$$d = \$219.05$$

8. A gym membership costs \$30 a month and personal training sessions cost extra. If a member's monthly bill is \$170, how much did they spend on personal training sessions?

$$30 + x = 170 \quad \$140 \text{ was spent on personal training sessions.}$$

$$x = 140$$

9. A custom bench was delivered and turned out to be too long. 4 inches were removed from the bench to make the new length 36 inches in length. What was the original length of the bench, in inches, when it was delivered?

$$b - 4 = 36 \quad \text{The bench was originally 40 in. long.}$$

$$b = 40$$

10. Sarah was sharing a story about her favorite birthday party from 8 years ago. If Sarah is now 21, how old was she at her favorite birthday party?

$$21 - b = 8 \quad \text{She was 13 at her favorite birthday party.}$$

$$b = 13$$

Grade 6 Accelerated
Unit 14
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

1. What is the inverse of multiplication?

division

2. What is the inverse of division?

multiplication

For questions 3-7, use the algebra tiles shown.



3. What expression represents the left side of the equation?

$\frac{x}{6}$ or $x \div 6$ or $\frac{1}{6}x$

4. What expression represents the right side of the equation?

3

5. Write the algebraic representation of the equation.

$\frac{x}{6} = 3$ or $x \div 6 = 3$ or $\frac{1}{6}x = 3$

6. What operation would be used to solve the equation?

multiplication

7. What value of x makes the equation true?

$x = 3(6)$

$x = 18$

$$8. \quad 11 \cdot -4 = \frac{f}{11} \cdot 11$$

$$f = -44$$

$$9. \quad 2 \cdot -6 = \frac{y}{2} \cdot 2$$

$$y = -12$$

$$10. \quad \frac{-14x}{-14} = \frac{-56}{-14}$$

$$x = 4$$

Grade 6 Accelerated
Unit 14
Lesson 7 Homework

Name Answer Key
Date _____
Period _____

Solve.

1. $-5 \cdot 25 = \frac{j}{-5} \cdot -5$

$$j = -125$$

2. $\frac{6x}{6} = \frac{42}{6}$

$$x = 7$$

3. $5 \cdot \frac{h}{5} = 6 \cdot 5$

$$h = 30$$

4. $\frac{-440}{44} = \frac{44m}{44}$

$$m = -10$$

5. $\frac{-34}{2} = \frac{2x}{2}$

$$x = -17$$

6. $-9 \cdot 21 = -\frac{y}{9} \cdot -9$

$$y = -189$$

$$7. \quad 4 \cdot \frac{r}{4} = 72 \cdot 4$$

$$r = 288$$

$$8. \quad \frac{-12p}{-12} = \frac{36}{-12}$$

$$p = -3$$

$$9. \quad 5 \cdot -3 = \frac{y}{5} \cdot 5$$

$$y = -15$$

$$10. \quad \frac{-x}{-1} = \frac{56}{-1}$$

$$x = -56$$

Grade 6 Accelerated
Unit 14
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

Write and solve an algebraic equation for each of the following.

1. At a restaurant, 6 friends decided to split the bill evenly. If each person paid \$14, what was the total bill?

$$\frac{t}{6} = \$14 \quad t = \$84 \quad \text{The total bill was \$84.}$$

2. A Great White Shark has 300 teeth. If the teeth are evenly arranged into 3 rows of teeth, how many teeth are in each row?

$$\frac{300}{t} = 3 \quad t = 100 \quad \text{Each row has 100 teeth.}$$

3. A school has a budget of \$400,000 for support staff members. If each support staff member makes \$40,000 each, how many support staff members can the school employ?

$$\frac{40,000s}{40,000} = \frac{400,000}{40,000} \quad \text{The school can employ 10 support staff members.}$$
$$s = 10$$

4. During a math adoption, there were 4 textbook company finalists invited to present their materials to the school district. For spacing purposes, each company could only bring a maximum of 3 people each. What is the maximum amount of people that could attend from the textbook companies?

$$4 \cdot 3 = t \text{ or } \frac{t}{4} = 3 \quad 12 \text{ people max can attend.}$$
$$t = 12$$

5. A princess dress takes 6 yards of fabric to create. If an order is placed for 9 dresses, how many yards of fabric will be needed?

$$\frac{d}{9} = 6 \quad d = 54 \quad 54 \text{ yards of fabric will be needed.}$$

6. A certain apple tree can produce 210 apples at a given time. If the apples are placed into bins that can hold 21 apples each, how many crates will be needed for the entire tree?

$$\frac{210}{c} = 21 \quad c = 10 \quad 10 \text{ crates will be needed.}$$

7. A recipe for Red Velvet cake calls for 1.5 sticks of butter for each cake. If someone wants to make 4 cakes, how many sticks of butter will be needed?

$$\frac{b}{4} = 1.5 \quad b = 6 \quad 6 \text{ sticks of butter will be needed.}$$

8. A new home builder offers 6 different floorplans, each that come with 5 color pallet options per floorplan. What is the number of different options that a buyer has with this builder?

$$\frac{f}{5} = 6 \quad f = 30 \quad \text{Buyers have 30 different options.}$$

9. A garment moving box can hold 8 suit jackets. If 4 garment boxes can fit in the back of a vehicle, what is the maximum number of suit jackets that can be moved in one trip?

$$\frac{g}{4} = 8 \quad g = 32 \quad 32 \text{ suit jackets can be moved.}$$

10. Thelma is participating in a Walk-a-Thon where she will get \$0.25 per lap that she walks. How many laps would she need to walk to reach a goal of \$50?

$$0.25w = 50 \quad \text{Thelma needs to walk 200 laps.}$$
$$w = 200$$

Grade 6 Accelerated
Unit 14
Lesson 9 Homework

Name Answer Key
Date _____
Period _____

1. For each inequality, write three numbers that could represent the solution set.

Answer may vary.

A. $x > 9$ 10, 11, 12

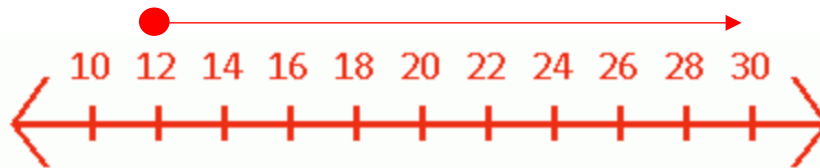
B. $-12 \leq m$ -11, -9, -7

C. $g < 0$ -1, -5, -10

2. The Overseas Highway, also known as the "Highway that Goes to Sea," runs 113 miles from Miami down to Key West. The road connects the Florida Keys to the Florida mainland. The speed limit of the road is 55 miles per hour. Represent the speeds that one could drive without exceeding the speed limit in an inequality with s representing the speed.

$s \leq 55$ mph

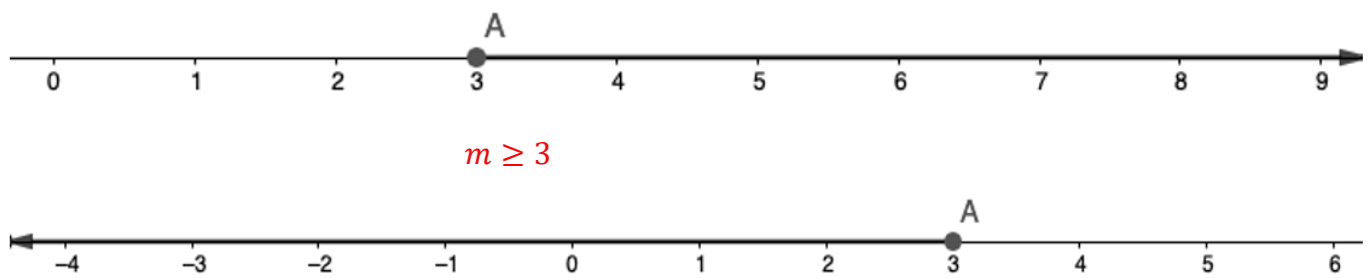
3. Graph the solution $x \geq 12$.



4. Chris said that the numbers $(-9, -10, -11)$ be in the solution set for the inequality $r < -9$. Explain if Chris' answer is correct.

-9 is incorrect because the inequality says r is "less than -9 , not "less than or equal to."

5. Compare and contrast the two number lines.



3 is part of the solution set of both inequalities.

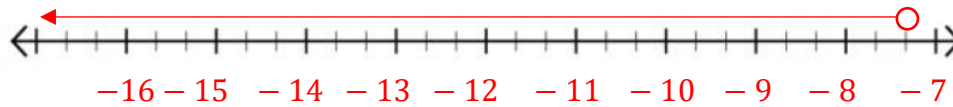
$m \leq 3$

6. Use the inequality $-9.75 > y - 2.5$ to complete the following:

A. Find the solution set by isolating the variable. Show your work.

$$\begin{array}{rcl} -9.75 & > & y - 2.5 \\ +2.5 & & +2.5 \\ \hline -7.25 & > & y \quad \text{or} \quad y < -7.25 \end{array}$$

B. Graph the solution set on the number line.



C. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

$$\begin{array}{ll} -9.75 > -10 - 2.5 & -9.75 > -16 - 2.5 \\ -9.75 > -12.5 \text{ T} & -9.75 > -18.5 \text{ T} \end{array}$$

7. How do you isolate a variable when solving an inequality?
Use the opposite operations just as you would when solving an equation.

Solve each inequality. Then, graph its solution set.

8. $-15\frac{1}{4} \leq m + 6.5$

$$\begin{array}{rcl} -6.5 & & -6.5 \\ -21.75 & \leq & m \\ m & \geq & -21.75 \end{array}$$



9. $48 > y - 24$

$$\begin{array}{rcl} +24 & & +24 \\ 72 & > & y \\ y & < & 72 \end{array}$$



10. $21.4 + w < 56.9$

$$\begin{array}{rcl} -21.4 & & -21.4 \\ w & < & 35.5 \end{array}$$



Grade 6 Accelerated
Unit 14
Lesson 10 Homework

Name Answer Key
Date _____
Period _____

1. When flying on an airplane, your travel bag can weigh no more than 50 lbs. Your bag currently weighs 27.3 lbs. The inequality $27.3 + m \leq 50$ represents the situation, where m represents the amount of weight that can be added to your bag. What is the maximum amount of weight that can be added to your bag while still making the weight requirement?

$$\begin{array}{r} 27.3 + m \leq 50 \\ -27.3 \quad -27.3 \\ \hline m \leq 22.7 \end{array}$$

No more than 22.7 lbs. can be added to your bag.

2. You earn \$8.75 per hour for walking dogs in your neighborhood. You are saving up your dog walking money to purchase a new kayak that costs \$280. The inequality $8.75h \geq 280$ represents the situation, where h represents the number of hours worked. What is the minimum number of hours you must walk in order to purchase the kayak?

$$\begin{array}{r} 8.75h \geq 280 \\ 8.75 \quad 8.75 \\ \hline h \geq 32 \end{array}$$

You must work at least 32 hours in order to purchase the kayak.

3. Complete the table by substituting each value of x to determine if that is a solution to each inequality.

Value of x	What is the value of $-4.5x$	Is $-4.5x > 18$ true or false?	Solution?
-4	18	$18 > 18$	☐ Yes ☐ No
-1.5	6.75	$6.75 > 18$	☐ Yes ☐ No
3	-13.5	$-13.5 > 18$	☐ Yes ☐ No
$6\frac{1}{5}$	-27.9	$-27.9 > 18$	☐ Yes ☐ No

4. Graph the inequality on the number line.



5. Find the solution to both inequalities using the division property of inequality.

$\frac{-7y}{-7} \leq \frac{-35}{-7}$ $y \geq 5$	$\frac{7y}{7} \leq \frac{-35}{7}$ $y \leq -5$
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6. Find the solution to both inequalities using the multiplication property of inequality.

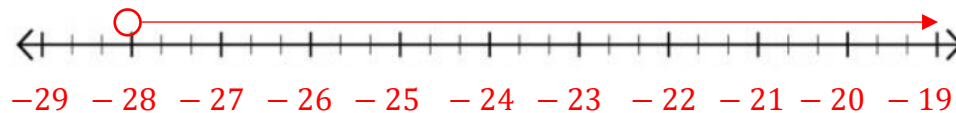
$\frac{y}{3} > 2$ $3 \cdot \frac{y}{3} > 2 \cdot 3$ $y > 6$	$\frac{y}{-3} > 2$ $-3 \cdot \frac{y}{-3} > 2 \cdot -3$ $y < -6$
---	--

Solve each inequality. Then, graph its solution set.

7. $\frac{r}{-4} < 7$

$$\frac{-4 \cdot \frac{r}{-4}}{-4} < \frac{7 \cdot -4}{-4}$$

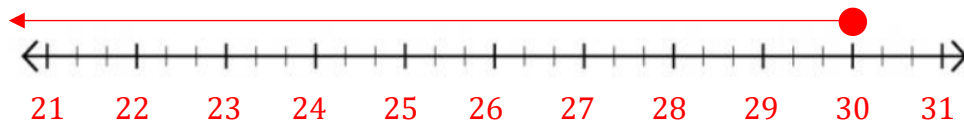
$$r > -28$$



8. $-0.4t \geq -12$

$$\frac{-0.4t}{-0.4} \geq \frac{-12}{-0.4}$$

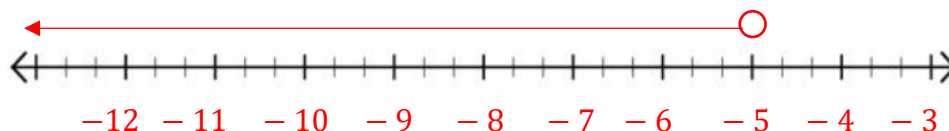
$$t \leq 30$$



9. $\frac{m}{-2} > \frac{5}{2}$

$$-2 \cdot \frac{m}{-2} > \frac{5}{2} \cdot -2$$

$$m < -5$$



10. Explain why the direction of the inequality symbol must be reversed when multiplying or dividing by the same negative number.

Because when you multiply or divide by a negative number, the value becomes its opposite (sign wise) and so the opposite sign is used.